A singleton first Betti support

A disproof of TLMC conjecture 00000000506

Complete self-contained Lean 4 formalization and mathematical report

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1. Claim, convention, and counterexample

The conjecture asserts, in both its English and Chinese versions, that the support of the first Betti row of a monomial ideal is connected and has no isolated points. It adds a proposed explanation involving the Taylor resolution. We refute the explicit no-isolated-points clause. No interpretation or verification of the additional Taylor-resolution clause is needed to refute the conjunction.

We use the standard total grading and the Betti numbers of the ideal as a module, not the Hilbert function of its entire syzygy module. For a minimal graded presentation $F_0 \twoheadrightarrow I$ with kernel K, the first Betti support is

$$\{j \in \mathbb{Z} : (K/\mathfrak{m}K)_j \neq 0\}.$$

This is the minimal-generator definition of the graded Betti row; it agrees with the shifts in a minimal graded free resolution and with graded Tor dimensions. See the definitions in [2, 3]. The formal proof computes this quotient explicitly rather than assigning a numerical table.

Theorem 1. *Let $R = \mathbb{F}_2[X, Y]$, with $\deg X = \deg Y = 1$, and let $I = (X, Y) = \mathfrak{m}$. Then I has minimal graded free resolution*

$$0 \longrightarrow R(-2) \xrightarrow{d_1} R(-1)^2 \xrightarrow{d_0} I \longrightarrow 0, \qquad d_1(h) = (Yh, Xh), \quad d_0(a, b) = Xa + Yb.$$

Consequently $\beta_{1,2}(I) = 1$ and $\beta_{1,j}(I) = 0$ for $j \neq 2$. Its first Betti support is the singleton $\{2\}$, which has an isolated point.

The field $\mathbb{F}_2$ is allowed: the source imposes no characteristic restriction. In characteristic two the usual Koszul relation $(-Y, X)$ is (Y, X). The ideal is genuinely monomial, generated by the monomials X and Y.

2. Exactness, with every syzygy accounted for

Represent a polynomial by its finitely supported coefficients $p_{ij} \in \mathbb{F}_2$ for $i, j \geq 0$. Multiplication by X or Y shifts the respective index by one, with zero on the new boundary.

Surjectivity of d_0 follows from the definition of (X, Y). Also $d_0 d_1(h) = XYh + YXh = 0$. To compute the entire kernel, suppose $Xa + Yb = 0$. Comparing coefficients gives

$$a_{i,0} = 0, \qquad b_{0,j} = 0, \qquad a_{i,j+1} = b_{i+1,j} \quad (i, j \geq 0).$$

Define $h_{ij} = a_{i,j+1}$. This still has finite support. The displayed identities yield $a = Yh$ and $b = Xh$. Conversely every such pair is a syzygy. The coefficient identity $(Yh)_{i,j+1} = h_{ij}$ proves uniqueness of h and injectivity of d_1. Thus the sequence is exact, including at its left end.

3. Minimality and the actual graded quotient

3.1 The original presentation is minimal

A polynomial lies in I if and only if its constant coefficient is zero. Indeed, split the terms with positive X exponent from those with X exponent zero and positive Y exponent. Thus $p \in I$ precisely when $p = Xa + Yb$.

Similarly,

$$p \in \mathfrak{m}I = XI + YI \quad \Longleftrightarrow \quad p_{00} = p_{10} = p_{01} = 0.$$

For the reverse implication the same term split gives coefficients a, b whose constant coefficients are p_{10}, p_{01}, hence $a, b \in I$. It follows that

$$I/\mathfrak{m}I \xrightarrow{\sim} \mathbb{F}_2^2, \qquad [p] \longmapsto (p_{10}, p_{01}).$$

The classes of X, Y are therefore a basis, both in degree one. This verifies minimality of the generators used to define K, a necessary point when computing first Betti numbers from a presentation.

3.2 Minimality of the differential

The map d_1 is R-linear and additive; d_0 is also R-linear and additive. The entries Y, X of d_1 lie in $\mathfrak{m}$. Thus the image of the free differential lies in $\mathfrak{m}R(-1)^2$, the standard minimal-complex condition. There is no further differential because d_1 is injective.

The grading is not implicit: if h is homogeneous of ordinary degree r, then (Yh, Xh) has degree $r + 2$ in $R(-1)^2$. This is exactly the shift $R(-2)$. The augmentation preserves degree. The ring multiplication is also proved to respect the standard total grading.

3.3 The quotient $K/\mathfrak{m}K$

The established R-linear bijection $d_1 : R(-2) \longrightarrow K$ takes $\mathfrak{m}$ onto $\mathfrak{m}K$. In detail, finite sums of arbitrary products of an element of $\mathfrak{m}$ with an element of K equal $XK + YK$, since $\mathfrak{m} = (X, Y)$. Under d_1, membership in this submodule is equivalent to $h \in \mathfrak{m}$, that is, $h_{00} = 0$.

Therefore the map

$$K/\mathfrak{m}K \longrightarrow \mathbb{F}_2, \qquad [d_1(h)] \longmapsto h_{00}$$

is a well-defined additive, $\mathbb{F}_2$-linear bijection. Its inverse sends $c \in \mathbb{F}_2$ to $[d_1(c)]$. In particular, the nonzero class of $d_1(1)$ is a one-element basis. This is a quotient of the actual polynomial syzygy module, not a separately invented two-element model.

Now a homogeneous syzygy of degree j can have nonzero quotient class only when $h_{00} = 1$. That constant coefficient contributes degree two, forcing $j = 2$. Conversely $d_1(1)$ is homogeneous of degree two and has nonzero class. This proves, for every integer j, including negative degrees,

$$(K/\mathfrak{m}K)_j \neq 0 \quad \Longleftrightarrow \quad j = 2.$$

The one-element basis also gives the stated numerical value $\beta_{1,2}(I) = 1$.

4. Why the singleton disproves the source

A singleton is connected. The argument does *not* claim otherwise. It contradicts the separate requirement that the support have no isolated points.

Proposition 1. *In every ambient topology, the sole point of a singleton subset is isolated in that subset. With any irreflexive adjacency relation, a one-vertex support also has no neighbor.*

Proof. The whole ambient space is open, and its intersection with the singleton is the singleton itself. Hence it is an isolating neighborhood. For adjacency, any putative neighbor in the support must be the same point, contradicting irreflexivity. □

Both statements are formalized. The topological theorem quantifies over an arbitrary open-set predicate, requiring only that the whole space is open; therefore it applies to any genuine topology. The graph theorem applies in particular to adjacency of consecutive integer degrees. This avoids relying on an unstated topological-versus-combinatorial convention in the source.

Alternative indexing: the quotient ring

Some authors tabulate the Betti numbers of R/I instead of I. In that convention the first relation module is I, and the first minimal-generator quotient is $I/\mathfrak{m}I$. Section 3.1 shows that it has two independent generators, both in degree one. Thus the alternative first support is $\{1\}$, again a singleton with an isolated point. The Lean source independently proves that the homogeneous nonzero-residue support of $I/\mathfrak{m}I$ is exactly $\{1\}$.

What is not being conflated

The Hilbert function of the *whole* syzygy module $K \cong R(-2)$ is different from its Betti row. It is nonzero in every degree $j \geq 2$. The conjecture explicitly names a Betti table, so the standard graded minimal-generator convention is used. The proof does not replace Betti support by the Hilbert support of K.

We do not need a general construction of Tor, the Taylor resolution, or a proof of uniqueness of every minimal free resolution. Instead, the exact minimal graded resolution and the defining minimal-generator quotient are verified directly for this particular monomial ideal. This is sufficient to refute the universal claim.

5. Formal model and scope

The project uses only Lean 4 core and Std, with no Mathlib dependency.

- A polynomial is a coefficient function $\mathbb{N}^2 \to \{0, 1\}$ with an existential finite-support bound. Equality ignores the chosen bound.
- Addition is coefficientwise exclusive-or. Multiplication is obtained from finite lists of monomials by adding exponents and collecting parity. Every finitely supported coefficient family has a list representation; representation independence and all commutative-ring laws are proved.
- The coefficient operations are proved to satisfy the two-element field laws. The free rank-two module has the actual polynomial scalar action.
- The ideal, kernels, finite-sum ideal products, grading, and quotient equivalence relation are explicit. The final support is not postulated.

6. Proof map and reproducibility

The final theorem is `TLMC506.conjecture506_counterexample` in `Resolution.lean`. It bundles the monomial-ideal certificate, the verified minimal resolution, the integer-indexed support identity, and failure of the no-isolated clause.

Key supporting results are:

- `polynomial_ring_laws`, `free_module_laws`, and `homogeneous_mul`: actual ring, module, and grading structures.
- `kernel_free_rank_one`: unique polynomial parametrization of every syzygy; `presentation_minimal_basis`: the basis of $I/\mathfrak{m}I$.
- `maximal_times_syzygy_is_ideal_product`: identification with the genuine finite-sum ideal-module product $\mathfrak{m}K$.
- `quotient_basis`: unique coefficient for every class in $K/\mathfrak{m}K$; `first_betti_supportZ_iff`: exact support over all integer degrees.
- `first_betti_supportZ_graph_isolated`: the graph reading; `quotient_first_betti_supportZ_iff`: the alternative indexing.

Verification procedure

The toolchain is pinned to `leanprover/lean4:v4.31.0`. Run `./VERIFY.sh` with that toolchain available on `PATH`, or set `LEAN_BIN_DIR` to its `bin` directory. The script copies *only authored sources and project configuration* into a fresh temporary directory, then performs:

1. a full Lake build and strict, warning-as-error source checks;
2. an environment-wide transitive axiom audit of every authored theorem, including generated supporting declarations;
3. official `leanchecker Main` kernel replay of the fresh build;
4. independent finite polynomial tests in `reproduce.py`.

The permitted axiom set is exactly Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. There are no added axioms, unfinished proofs, native evaluation proof escapes, or kernel bypasses. The Python checks are supplementary; no finite experiment is used to establish the universal theorem. Verification logs and a SHA-256 manifest are included in the package.

Repository eligibility and provenance

The dated read-only eligibility check used upstream commit `b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`. The exact metadata row for 00000000506 has both solved flags set to `false`. The source blob is `6a9833bfdd4546f0e5065e9a5bdaad0680da0c14`. The solution directory was absent, and padded, unpadded, semantic, and complete saved PR title/body checks found no relevant prior submission for this ID. A dated public eligibility record is included. This is a dated repository-status check, not a claim of worldwide mathematical novelty or a guarantee about later submissions. The elementary Koszul counterexample uses standard algebra.

References

[1] The Last Math Competition, Conjecture 00000000506, English and Chinese source, accessed 21 October 2026.

[2] J. Madden, *Lecture 7: Betti Numbers*, 2006, pp. 1–2. Primary lecture notes.

[3] D. Eisenbud, *The Geometry of Syzygies*, Springer, 2005, Section 1B, especially minimal resolutions and Corollary 1.5. Author's book, hosted course copy.